

PART 1 — LEVEL 1 (Basic / Board Pattern) ~150 Questions

SECTION 1: Amorphous & Crystalline Solids (Q1–20)

1. Define crystalline solids.
2. Define amorphous solids.
3. Give one example each of crystalline and amorphous solids.
4. Difference between crystalline and amorphous solids in terms of long-range order.
5. What is isotropy? Which solids show isotropic properties?
6. Define anisotropy. Give example.
7. Explain how melting points differ for crystalline and amorphous solids.
8. Give examples of amorphous solids used in everyday life.
9. Define polycrystalline solids.
10. Explain how glasses are formed.
11. Define allotropy. Give example.
12. Difference between single crystal and polycrystalline solids.
13. Which solids show sharp melting points?
14. What is short-range order? Which type of solid exhibits it?
15. Define unit cell in crystalline solids.
16. Give examples of crystalline solids with metallic bonding.
17. Give examples of crystalline solids with ionic bonding.
18. Give examples of crystalline solids with covalent bonding.
19. Explain molecular solids with example.
20. What is the coordination number in ionic solids?

SECTION 2: Crystal Lattices & Unit Cells (Q21–50)

21. Define crystal lattice.
22. Define lattice points.
23. Differentiate between lattice and motif.
24. What is primitive unit cell?
25. What is body-centered cubic (BCC) unit cell?
26. What is face-centered cubic (FCC) unit cell?
27. Give examples of BCC metals.
28. Give examples of FCC metals.
29. Define simple cubic (SC) unit cell.
30. Number of atoms in SC, BCC, FCC unit cells.
31. Edge length in terms of atomic radius for SC.
32. Edge length in terms of atomic radius for BCC.
33. Edge length in terms of atomic radius for FCC.

34. Define packing efficiency.
35. Packing efficiency for SC, BCC, FCC, HCP.
36. Difference between primitive and non-primitive unit cells.
37. Define close-packed structure.
38. Examples of metals with HCP structure.
39. Relationship between density, atomic mass, and unit cell dimensions.
40. Concept of lattice constant.

◆ SECTION 3: Number of Atoms in Unit Cell (Q51–70)

41. Number of atoms in SC unit cell.
42. Number of atoms in BCC unit cell.
43. Number of atoms in FCC unit cell.
44. Number of atoms in HCP unit cell.
45. Derive number of atoms in SC unit cell.
46. Derive number of atoms in BCC unit cell.
47. Derive number of atoms in FCC unit cell.
48. Derive number of atoms in HCP unit cell.
49. Fractional contribution of corner atom in a unit cell.
50. Fractional contribution of face-centered atom.
51. Fractional contribution of body-centered atom.
52. Fractional contribution of edge atom.
53. How many lattice points in simple cubic unit cell?
54. How many lattice points in BCC unit cell?
55. How many lattice points in FCC unit cell?
56. How many lattice points in HCP unit cell?
57. Calculate number of atoms per unit volume for FCC metal.
58. Calculate number of atoms per unit volume for BCC metal.
59. Differentiate atomic packing factor and packing efficiency.
60. Examples of metals with SC structure.

◆ SECTION 4: Packing Efficiency & Voids (Q71–100)

61. Define packing efficiency.
62. Packing efficiency of SC.
63. Packing efficiency of BCC.
64. Packing efficiency of FCC.
65. Packing efficiency of HCP.
66. Why SC has lowest packing efficiency?
67. Define void in crystal structure.
68. Voids in SC, BCC, FCC, HCP.

69. Tetrahedral voids: definition and examples.
70. Octahedral voids: definition and examples.
71. Fraction of tetrahedral voids per atom in FCC.
72. Fraction of octahedral voids per atom in FCC.
73. Fraction of tetrahedral voids per atom in BCC.
74. Fraction of octahedral voids per atom in BCC.
75. Explain why HCP and FCC have same packing efficiency.
76. Relation between radius of atom and size of tetrahedral void.
77. Relation between radius of atom and size of octahedral void.
78. Calculate radius of tetrahedral void in terms of atomic radius.
79. Calculate radius of octahedral void in terms of atomic radius.
80. Difference between interstitial site and void.

◆ SECTION 5: Density of Unit Cell (Q101–120)

81. Derive formula for density of unit cell.
82. Density in terms of atomic mass, number of atoms, and volume.
83. Calculate density of SC metal with given atomic radius and mass.
84. Calculate density of BCC metal.
85. Calculate density of FCC metal.
86. Explain why FCC metals are denser than SC metals.
87. Relation between atomic packing factor and density.
88. Calculate density for HCP structure.
89. Concept of molar volume in solids.
90. Difference between theoretical and experimental density.
91. Effect of voids on density of solid.
92. Density calculation of ionic solids using unit cell.
93. Calculate lattice parameter from density and atomic mass.
94. Convert atomic radius to edge length.
95. Calculate number of formula units in unit cell from density.
96. Calculate atomic radius from density of FCC metal.
97. Calculate atomic radius from density of BCC metal.
98. Compare densities of FCC and BCC metals of same atomic mass.
99. Density of compound AB in FCC structure.
100. Density of compound AB in BCC structure.

◆ SECTION 6: Imperfections in Solids (Q121–140)

101. Define point defects in solids.
102. Examples of point defects.
103. Schottky defect: definition and example.

- 104. Frenkel defect: definition and example.
- 105. Effect of Schottky defect on density.
- 106. Effect of Frenkel defect on density.
- 107. Line defects: definition.
- 108. Edge dislocation: definition.
- 109. Screw dislocation: definition.
- 110. Effects of dislocations on mechanical properties.
- 111. Surface defects: definition.
- 112. Grain boundaries: definition.
- 113. Stacking faults: definition.
- 114. Colour centres: definition and examples.
- 115. Defects in ionic solids: examples.
- 116. Intrinsic vs extrinsic defects.
- 117. Imperfections in metals.
- 118. Imperfections in ceramics.
- 119. Imperfections in semiconductors.
- 120. Relationship between defect concentration and temperature.

◆ SECTION 7: Electrical & Magnetic Properties (Q141–150)

- 121. Define electrical conductivity in solids.
- 122. Examples of good conductors.
- 123. Examples of insulators.
- 124. Examples of semiconductors.
- 125. Define intrinsic semiconductors.
- 126. Define extrinsic semiconductors.
- 127. Difference between n-type and p-type semiconductors.
- 128. Factors affecting electrical conductivity.
- 129. Define magnetic susceptibility.
- 130. Types of magnetism: diamagnetism, paramagnetism, ferromagnetism.
- 131. Examples of diamagnetic materials.
- 132. Examples of paramagnetic materials.
- 133. Examples of ferromagnetic materials.
- 134. Effect of temperature on magnetic properties.
- 135. Curie temperature definition.
- 136. Magnetic domains: definition.
- 137. Magnetic hysteresis: definition.
- 138. Applications of magnetic materials.
- 139. Superconductors: definition and examples.
- 140. Critical temperature of superconductor.
- 141. Meissner effect: definition.

142. Applications of superconductors in MRI and maglev.
143. Electrical conductivity of metals vs semiconductors.
144. Hall effect: definition and applications.
145. Nernst effect: definition.
146. Piezoelectric solids: definition.
147. Examples of piezoelectric solids.
148. Relationship between crystal structure and electrical properties.
149. Effect of defects on conductivity.
150. Effect of impurities on magnetic properties.